Assignment 3

1. Information Presentation Response

Choose one of the information presentations, and in 3-5 sentences, respond to the following prompt:

- Has learning more about this topic changed your view on/opinion of it? If so, then what caused the change in the way you think/feel? If not, then what evidence could theoretically be presented that would persuade you to change your opinions?

2. Algorithmic University Admissions

You have recently been hired as a consultant for the completely fictional University of Scarborough, Toronto Campus in their admissions department. Until now, their admissions has been based on a very simple formula: sort the students by their highschool grade average, and offer placements to the top n students where n is determined by teaching capacity.

The reason you've been hired is to lead a team developing a new admissions program titled the Program for Optimizing Student Tuition (POST) that will (hopefully) be better at selecting high quality candidates. You have access to essentially unlimited information on prospective students, including not just high school marks, but also age, gender, race, extra-curricular activities, family income, address, parental education history, criminal history, and taxation history. Furthermore, you have 40 years worth of the same data on every student who has attended USTC, correlated with their GPA.

The admissions department thinks this should be a piece of cake. Just throw all the data into a machine learning model, and choose the students who are statistically most likely to succeed. But you have some reservations, having read about algorithmic bias in these sorts of systems.

Your job is to provide a short report of around 500 words advising your bosses on what (if any) of the data should NOT be included into your system. They will be sceptical of anything that would reduce the quality of the algorithm, so you'll need to give specific examples of the problems that certain types of data could cause, preferably with evidence.

3. Common Crawl

With the explosion of generative AI models in recent years, a lot of attention has been paid to the potential uses these models may be put to, as well as the challenges they could pose to various areas of society. A less common area of discussion is the corpora on which these models are built. Where they come from, how they're built, and who has access can have huge impacts on the models themselves.

Your job is to research the Common Crawl repository (`https://commoncrawl.org`) and write a report that summarizes in 500-700 words how the data is chosen and filtered, and how the biases in that data could impact the development of large language models.

Remember that your report is not meant to focus on the technicalities of the dataset itself, nor the details of the LLM models based on it, but rather how the biases and structures of the dataset is having an impact on the developing field of generative AI.